

Hamzeh Abdallah

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Summary

Full Stack Software Engineer building Python/FastAPI backends and Vue.js/React frontends on PostgreSQL. Built 50+ async REST endpoints across 30+ schemas and 40+ reusable UI components for a production-grade platform, and graduated summa cum laude with 3.92 GPA.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL (PostgreSQL, MySQL), HTML, CSS, PHP

Frontend: Vue.js, Quasar, Pinia, Next.js, React.js, React Native, Expo

Backend: FastAPI, Flask, Node.js, Tortoise ORM, Pydantic, RESTful APIs, OAuth, pytest, unit testing, PartyKit, WebSockets

DevOps and Tools: Git, GitHub, Docker, Selenium, GitHub Actions, CI/CD

AI/LLM Tools: OpenAI GPT API, Claude Code, LM Studio, Cursor

Work Experience

AYA Holdings Group | *May 2023 - Sep 2024*

Full Stack Software Engineer | Remote

- Architected 4 independent FastAPI microservices with isolated PostgreSQL schemas, Tortoise ORM, and Docker Compose.
- Designed and delivered 50+ async RESTful endpoints across 30+ normalized PostgreSQL schemas with Aerich migrations, indexed foreign keys, and Pydantic v2 response validation.
- Engineered a 5-level content hierarchy model with a 3-stage review workflow and role-based visibility controls across the content management system.
- Developed 40+ reusable Vue.js/Quasar components across multiple platform features with internationalization supporting 100+ locales and full RTL language layouts.
- Built an automated content ingestion pipeline with UUID-based deduplication and OpenAI GPT-4 summarization, and integrated AI-assisted content generation into the platform's authoring backend.

Al-Ghazaly High School | *Sep 2024 - Feb 2025*

Substitute Teacher | Wayne, NJ

- Delivered day-to-day substitute instruction across grade levels and subjects, including 1-2 week extended assignments covering High School Computer Science and Middle School Science and Math curricula.

Projects

Tunge - Online Card Game | *TypeScript, Next.js, React Native, PartyKit* | *Aug 2025 - Present*

- Building a real-time 2-player card game targeting web and mobile using strict-mode TypeScript Turborepo, featuring a shared game engine, covered by 26 Vitest tests, and 3D animations for web and mobile.
- Built the PartyKit WebSocket server as the authoritative source of truth, binding each connection to a player index, rejecting off-turn moves, and stamping the Tunge caller's identity server-side to block spoofing, while broadcasting per-connection payloads that hide each player's drawn card from the opponent.
- Resolved a turn-order race condition in the Tunge penalty mechanic by tracking the offending player explicitly in shared game state.

IdleSpiral - Desktop Automation Application | *Python, PyAutoGUI, CustomTkinter* | *Apr 2025 - Jun 2025*

- Automated a farming loop requiring manual input every 2-3 minutes, enabling unattended 6-hour overnight sessions completing 140+ battle cycles, using Pillow screen template matching (90% confidence threshold) and a threaded PyAutoGUI card-play engine.

Unlimited Drinks | *Vue.js, Quasar, FastAPI, Docker, PostgreSQL* | *Sep 2023 - Dec 2023*

<https://hamzehab.github.io/unlimiteddrinks/#/>

- Built a full-stack beverage e-commerce platform with 12 frontend pages and 6 REST API routers, featuring a Stripe webhook handler that atomically creates orders, decrements inventory, and computes sales tax - deployed to Render, Supabase, and GitHub Pages via GitHub Actions.
- Prevented inventory overselling using stock validation at both the client and server and enforced a one review per customer product with OAuth 2.0/Auth0 route guards and database uniqueness constraints.

Education

Montclair State University

Bachelor of Science in Computer Science | Montclair, NJ

- Cumulative GPA: 3.92/4.0
- Graduated *summa cum laude*
- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Database Systems, Web Development, Computer Networks and Security